The 10 largest designs: ground truth vs ours

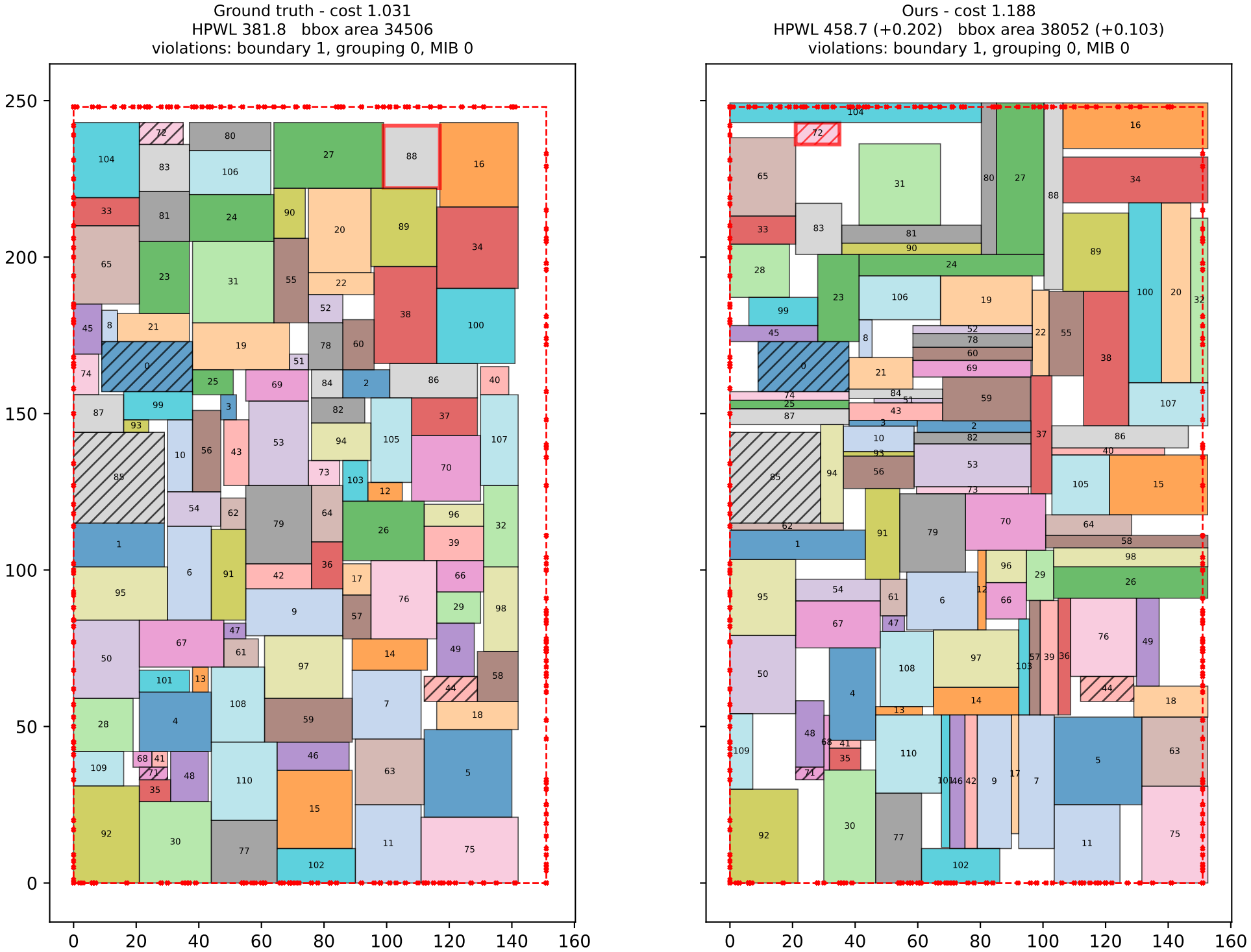
Our layouts from official_eval_1018_restart.json (10/18 official run). Contest cost, lower is better; the gaps are relative to the ground truth.

Design	Blocks	GT cost	Our cost	Result	HPWL gap	Area gap	Boundary viol. GT / ours	Grouping viol. GT / ours	Solve time
90	111	1.031	1.188	loss	+0.202	+0.103	1 / 1	0 / 0	2554 s
91	112	1.071	1.013	win	-0.001	+0.027	1 / 0	1 / 0	873 s
92	113	1.036	1.000	win	-0.021	-0.007	1 / 0	0 / 0	880 s
93	114	1.033	1.000	win	-0.009	-0.007	1 / 0	0 / 0	896 s
94	115	1.097	1.104	loss	+0.110	+0.032	3 / 1	0 / 0	2742 s
95	116	1.129	1.000	win	-0.008	-0.010	4 / 0	0 / 0	932 s
96	117	1.097	1.002	win	+0.004	-0.018	3 / 0	0 / 0	951 s
97	118	1.069	1.000	win	+0.000	-0.020	2 / 0	0 / 0	970 s
98	119	1.080	1.000	win	-0.011	-0.013	2 / 0	0 / 0	974 s
99	120	1.127	1.038	win	+0.003	+0.013	4 / 1	0 / 0	3007 s

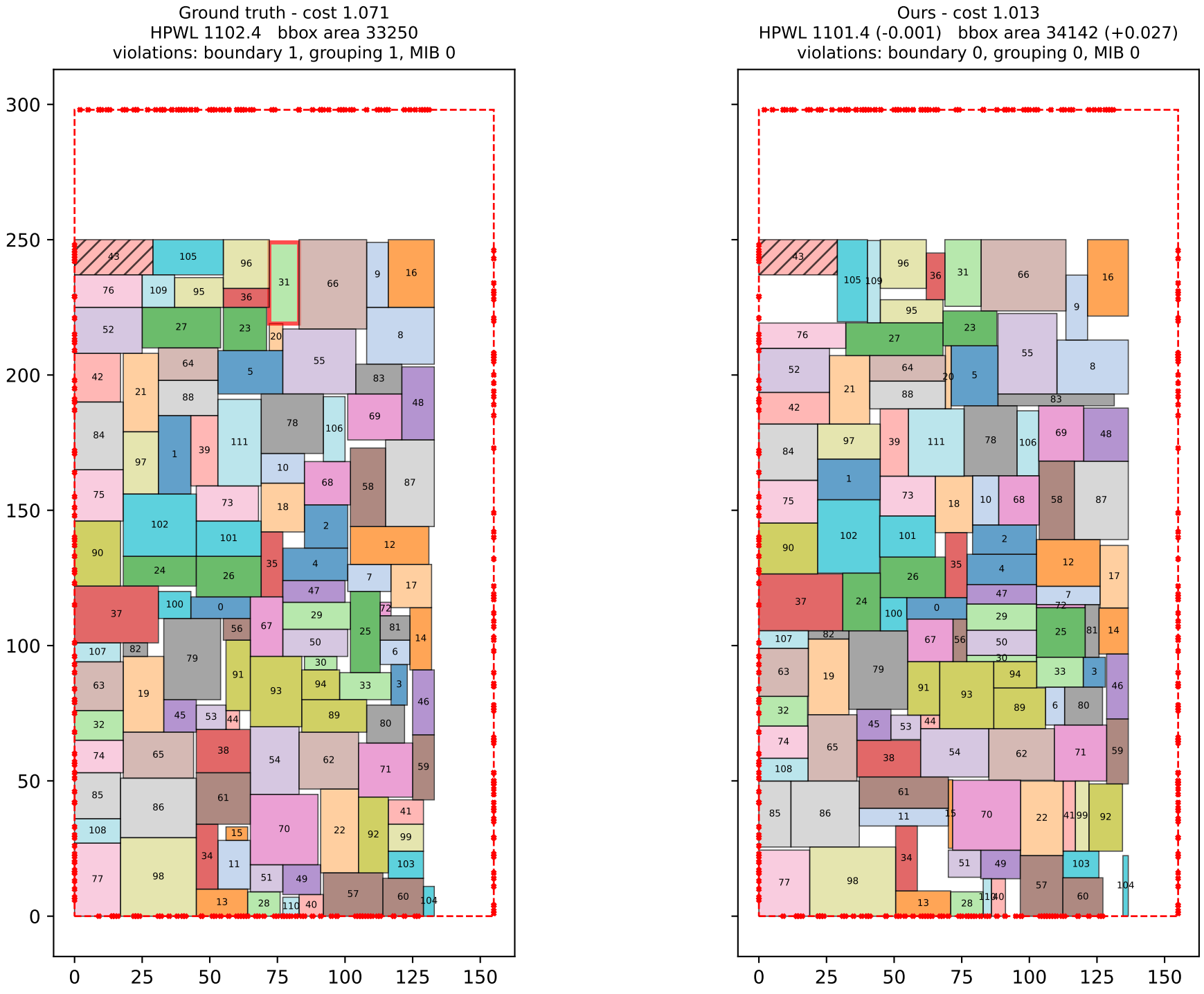
We beat the ground truth on 8 of 10. Weighted cost over these designs (contest weights e^(blocks/12)): ground truth 1.0822, ours 1.0289. These 10 designs carry 57% of the total contest weight.

Images: same axes for both layouts. Hatched = preplaced block; red outline = block that misses its required edge (boundary violation); red x = pins; dashed red = pin box. The ground truth also has boundary violations - the evaluator scores it the same way as ours.

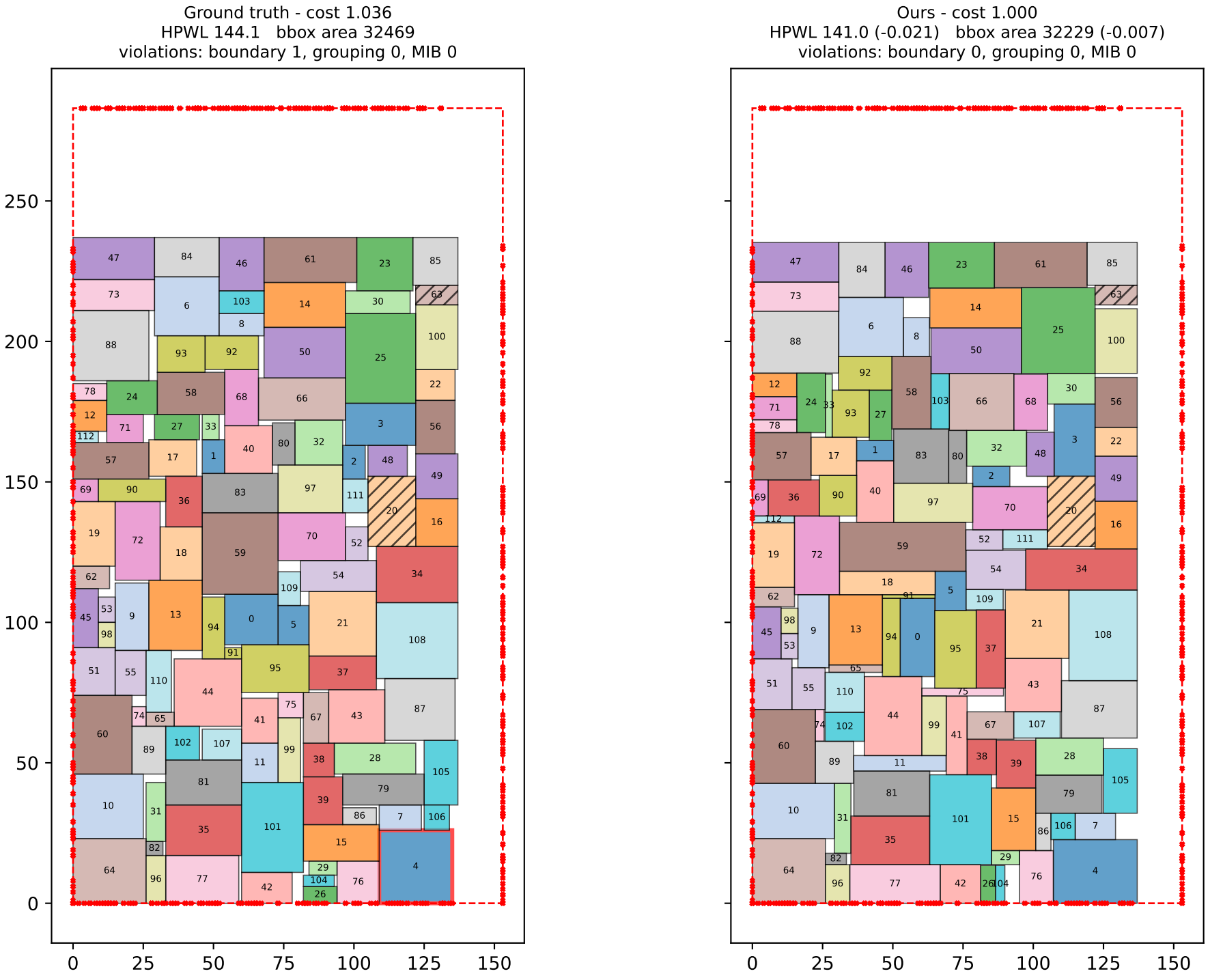
Design 90 (111 blocks): ours 1.188 vs ground truth 1.031 - ground truth wins



Design 91 (112 blocks): ours 1.013 vs ground truth 1.071 - we win



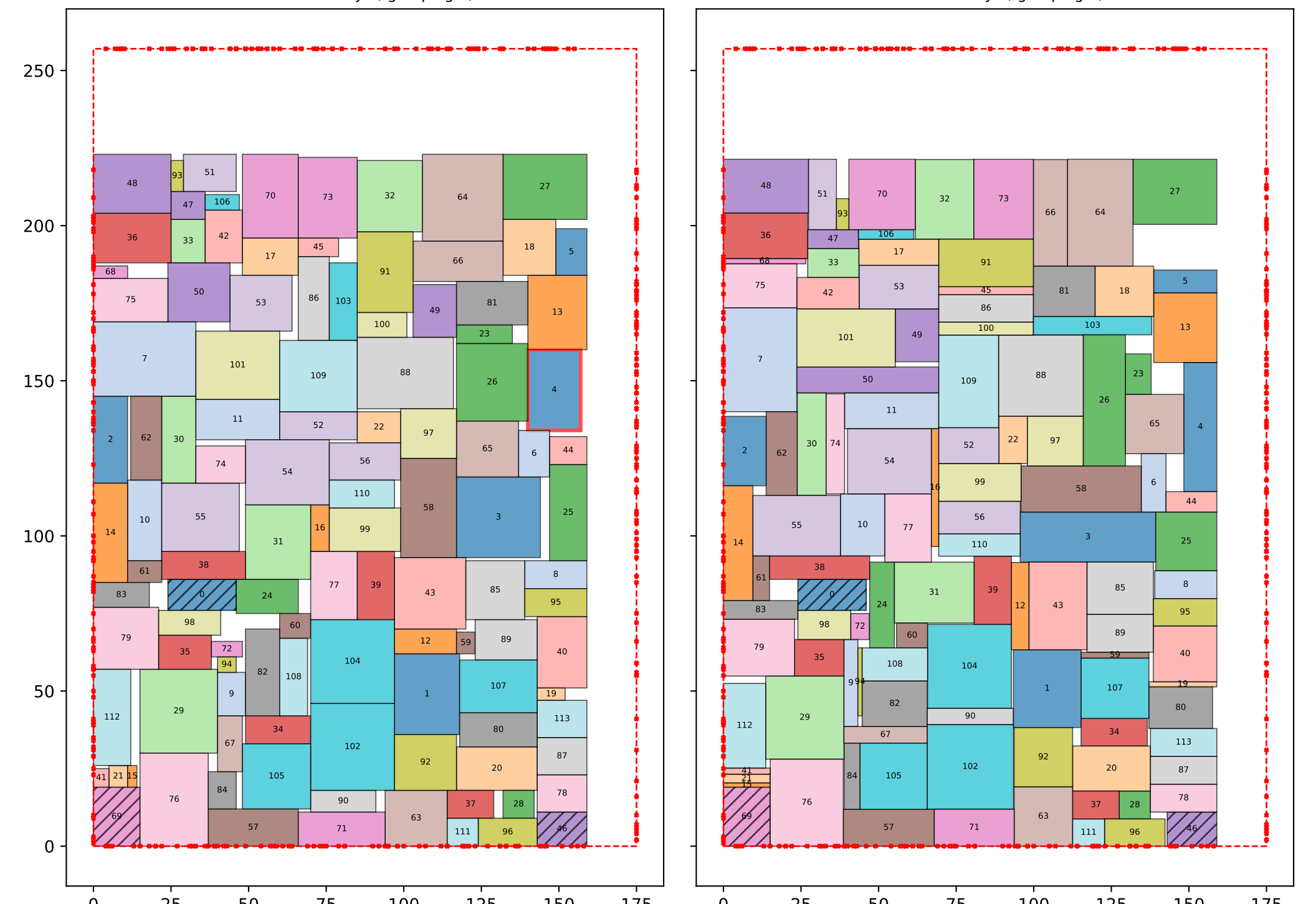
Design 92 (113 blocks): ours 1.000 vs ground truth 1.036 - we win



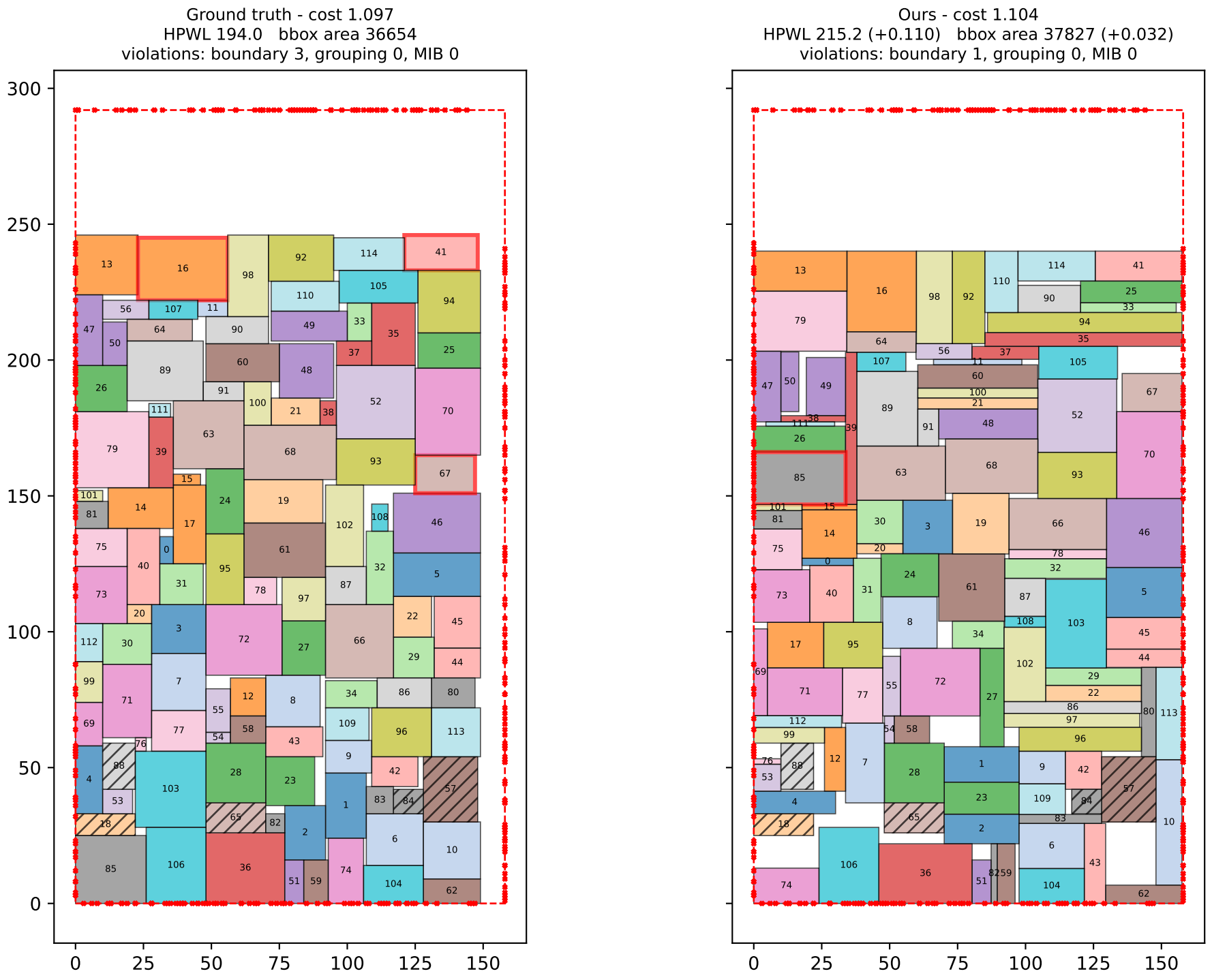
Design 93 (114 blocks): ours 1.000 vs ground truth 1.033 - we win

Ground truth - cost 1.033
HPWL 227.9 bbox area 35457
violations: boundary 1, grouping 0, MIB 0

Ours - cost 1.000
HPWL 225.9 (-0.009) bbox area 35201 (-0.007)
violations: boundary 0, grouping 0, MIB 0

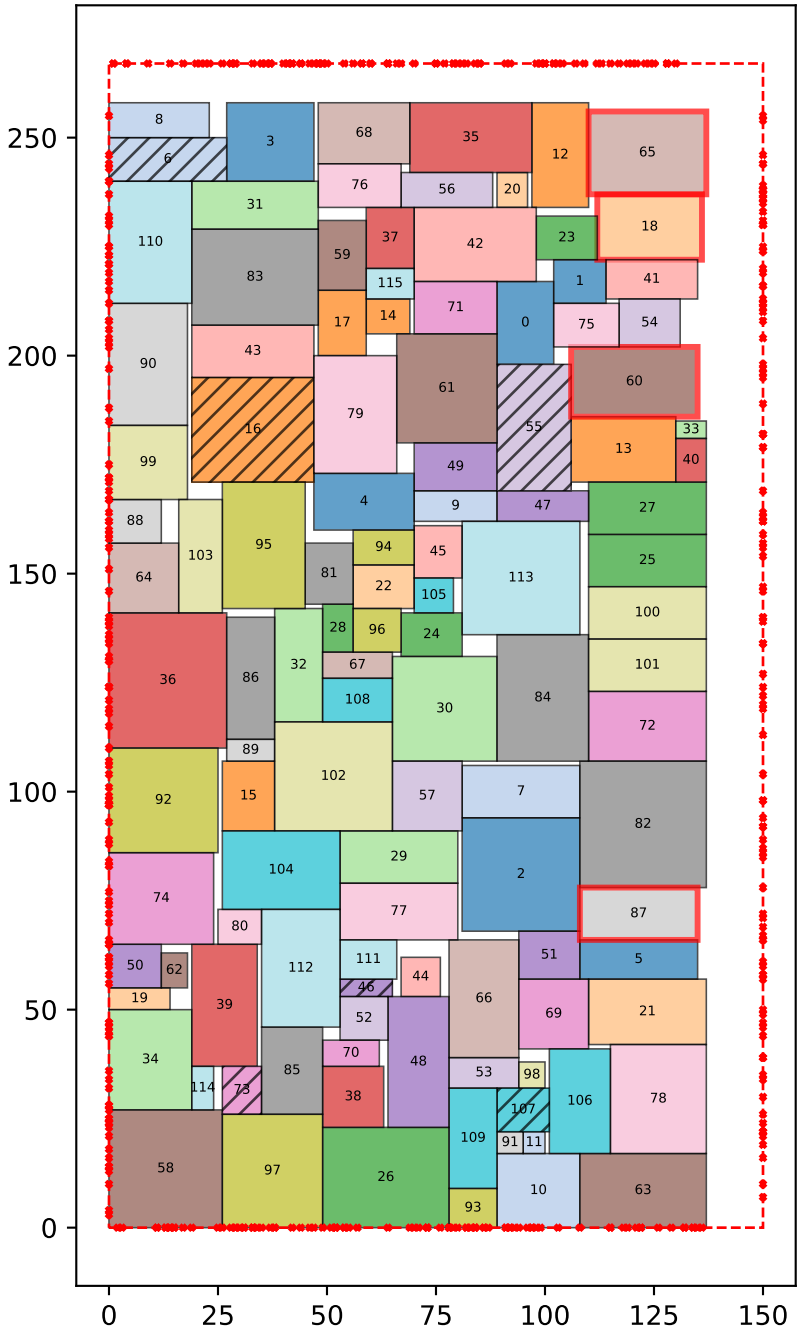


Design 94 (115 blocks): ours 1.104 vs ground truth 1.097 - ground truth wins

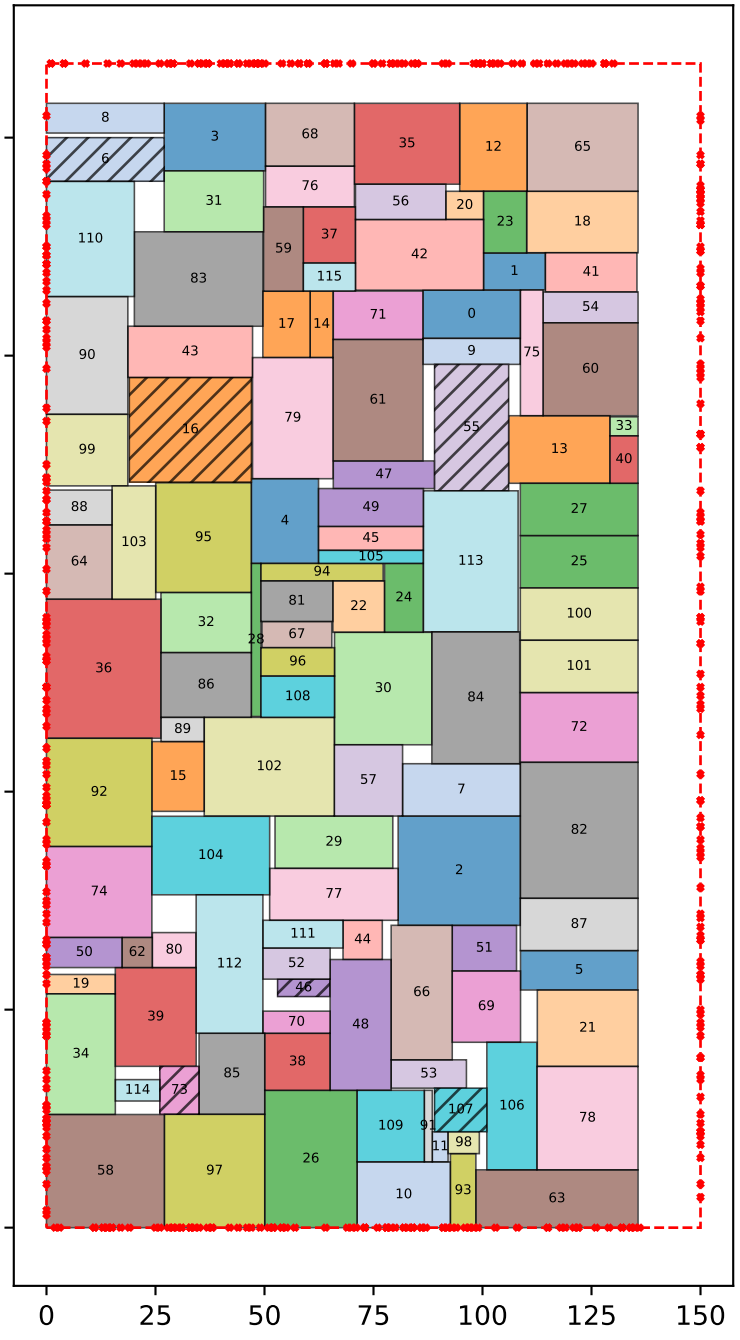


Design 95 (116 blocks): ours 1.000 vs ground truth 1.129 - we win

Ground truth - cost 1.129
HPWL 327.6 bbox area 35346
violations: boundary 4, grouping 0, MIB 0

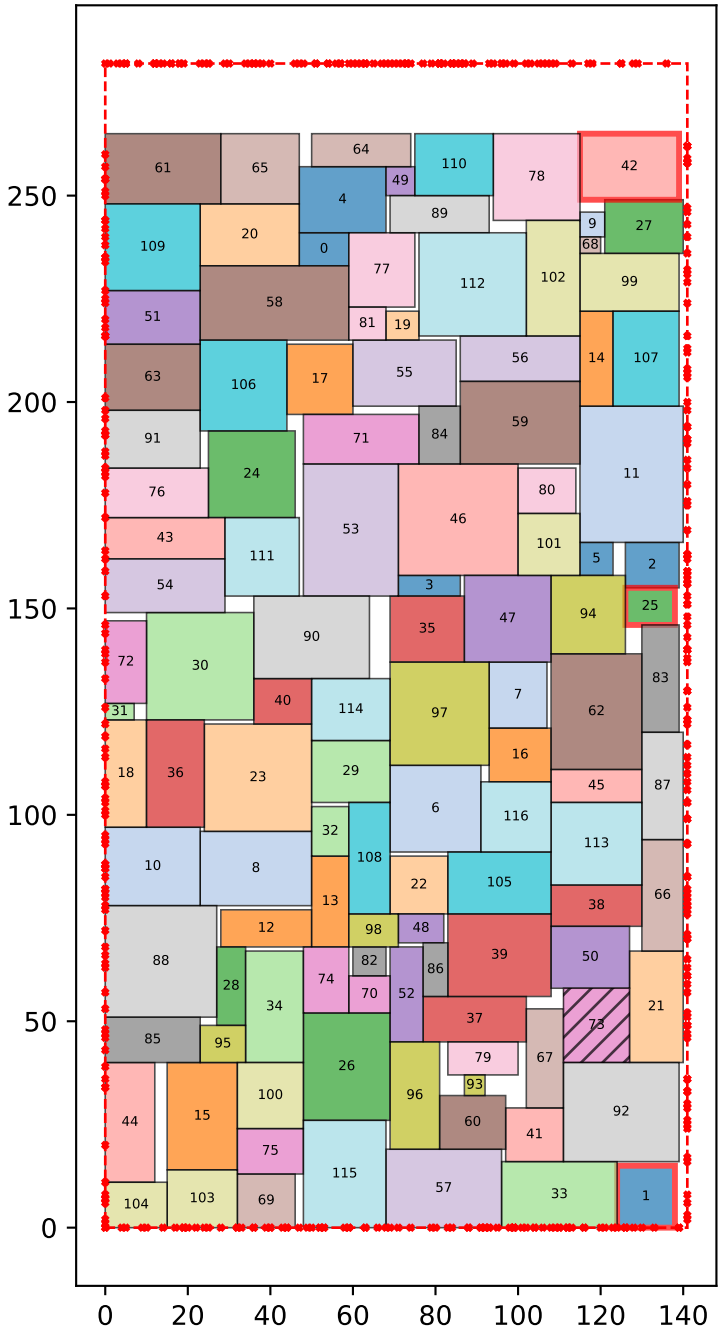


Ours - cost 1.000
HPWL 325.0 (-0.008) bbox area 34986 (-0.010)
violations: boundary 0, grouping 0, MIB 0

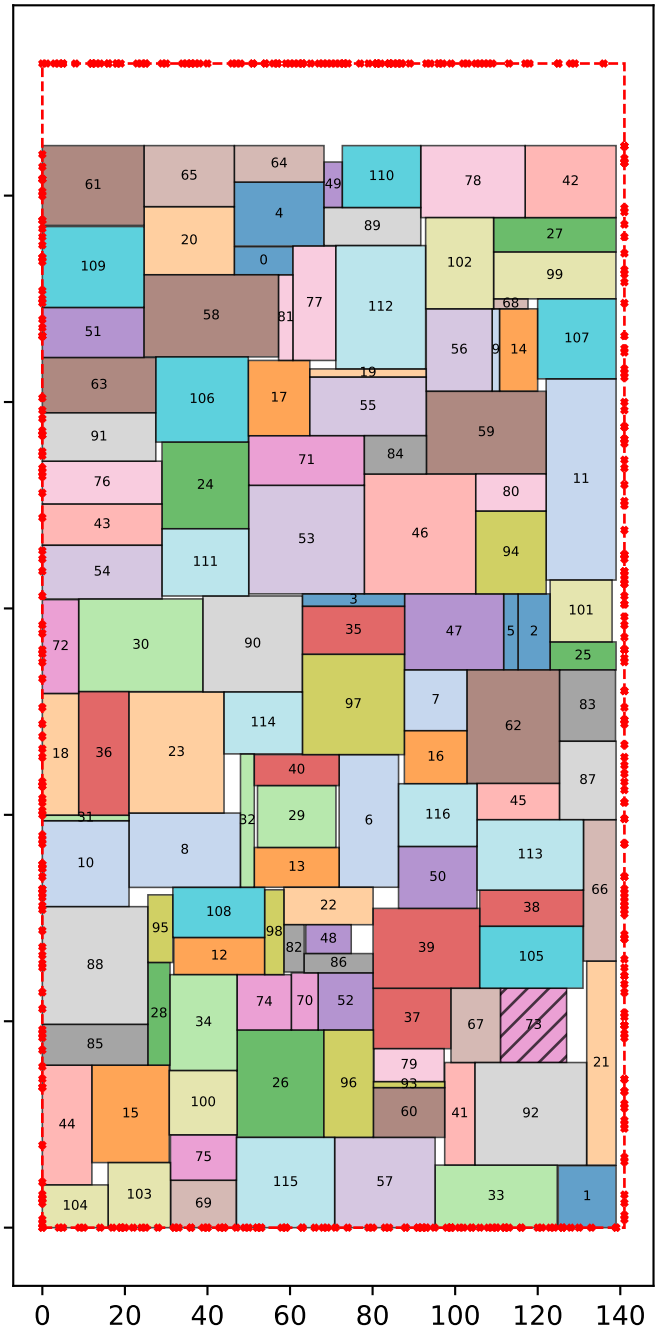


Design 96 (117 blocks): ours 1.002 vs ground truth 1.097 - we win

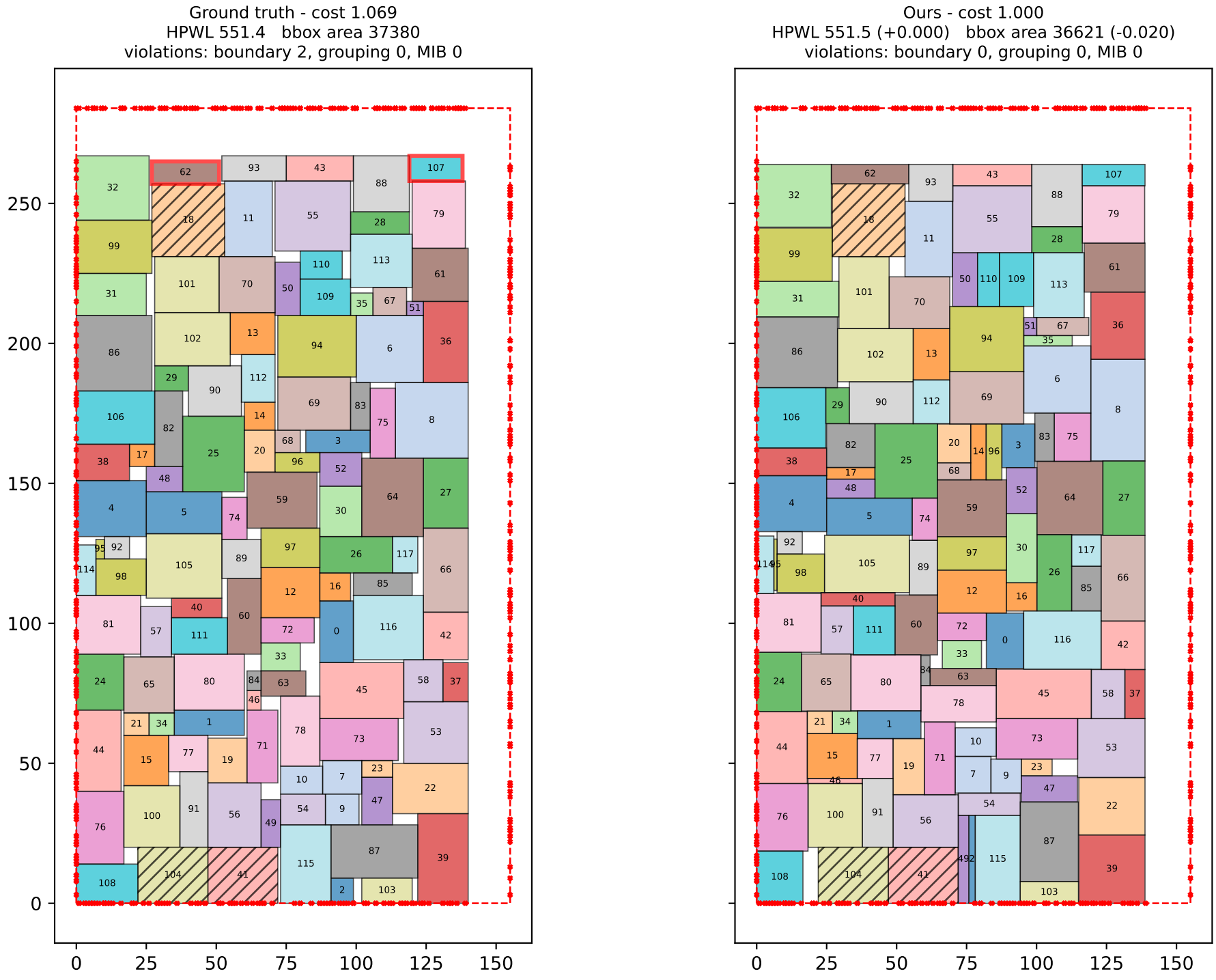
Ground truth - cost 1.097
HPWL 1100.6 bbox area 37100
violations: boundary 3, grouping 0, MIB 0



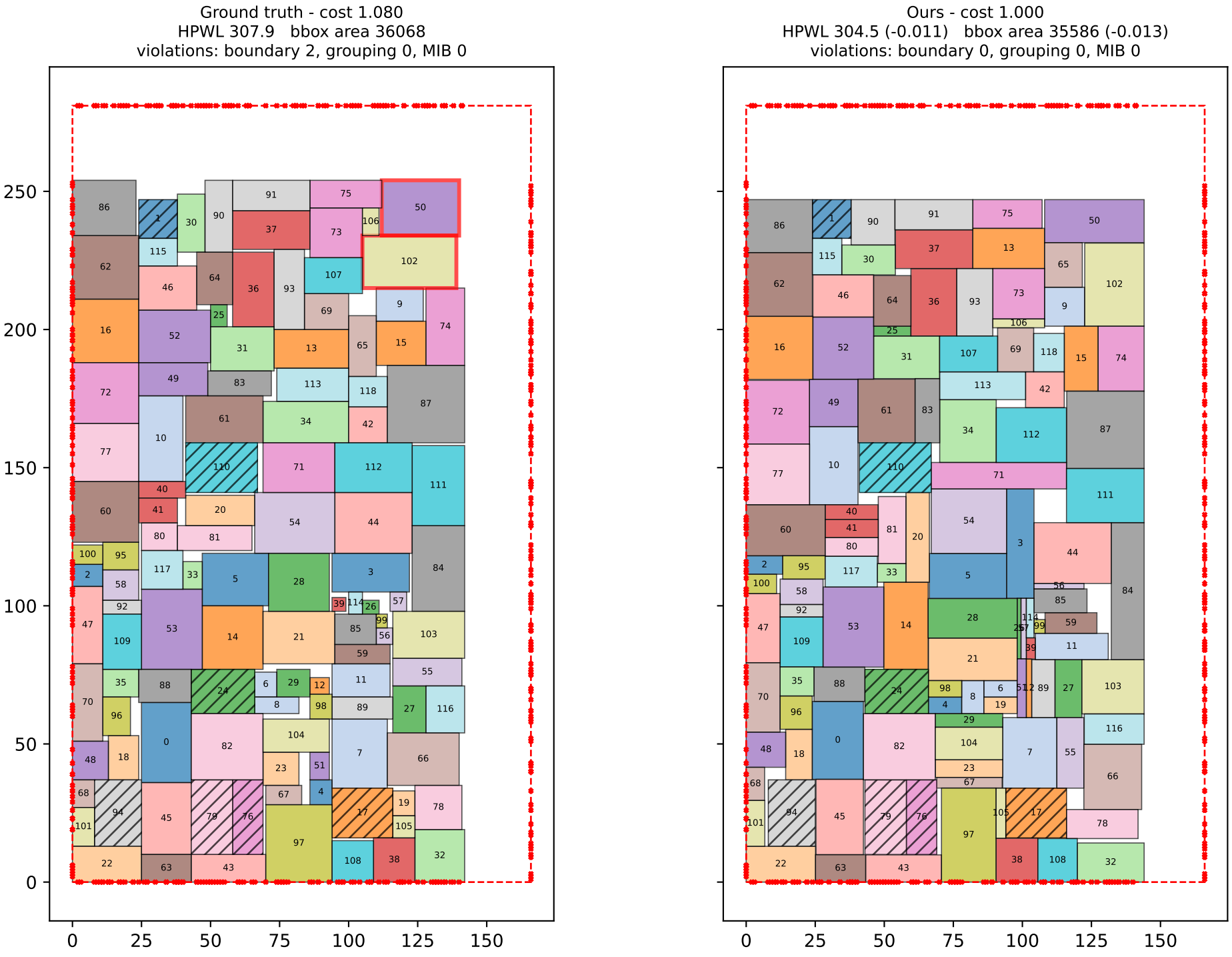
Ours - cost 1.002
HPWL 1105.2 (+0.004) bbox area 36427 (-0.018)
violations: boundary 0, grouping 0, MIB 0



Design 97 (118 blocks): ours 1.000 vs ground truth 1.069 - we win



Design 98 (119 blocks): ours 1.000 vs ground truth 1.080 - we win



Design 99 (120 blocks): ours 1.038 vs ground truth 1.127 - we win

